CS471 Lab 1

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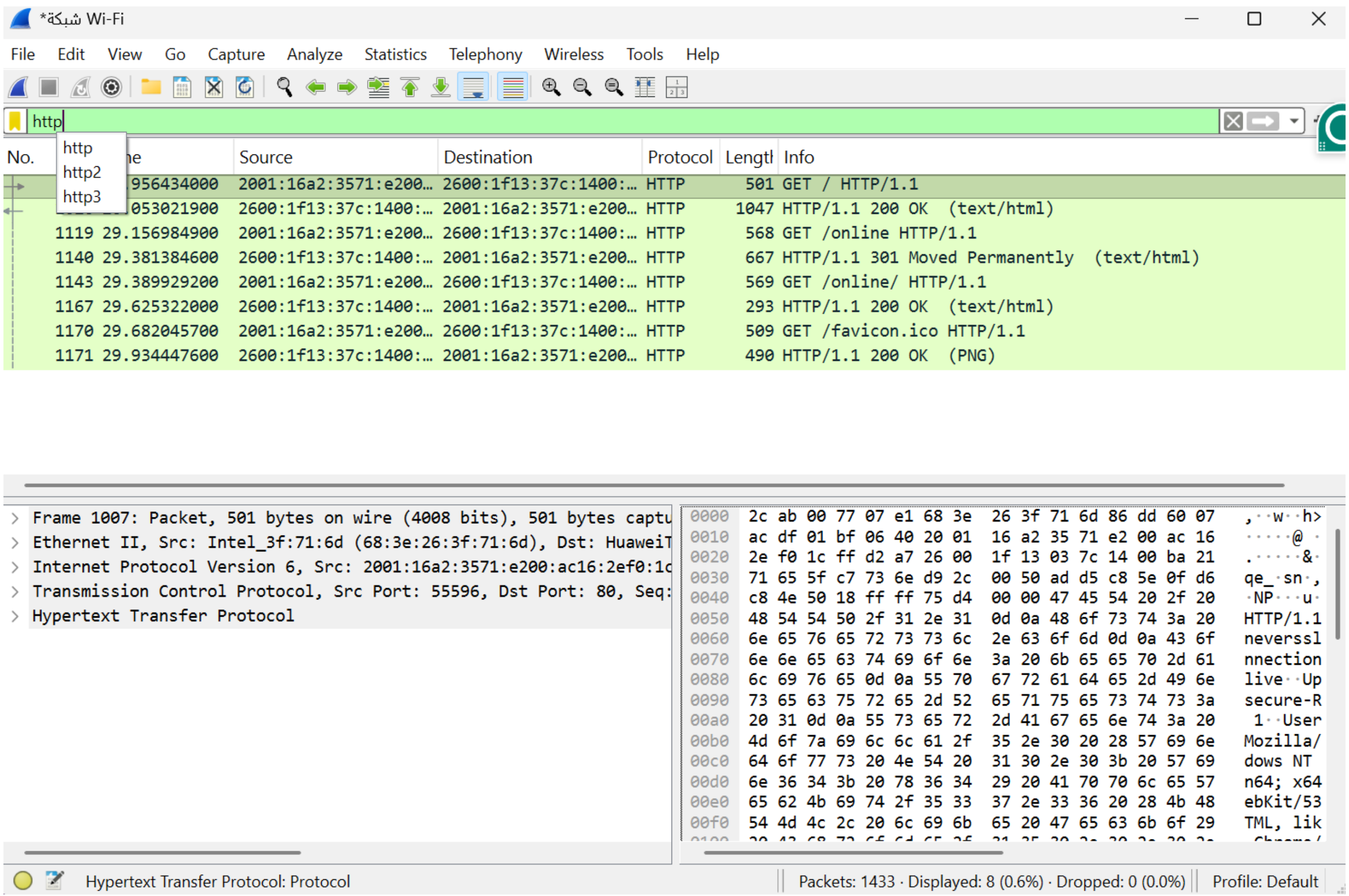
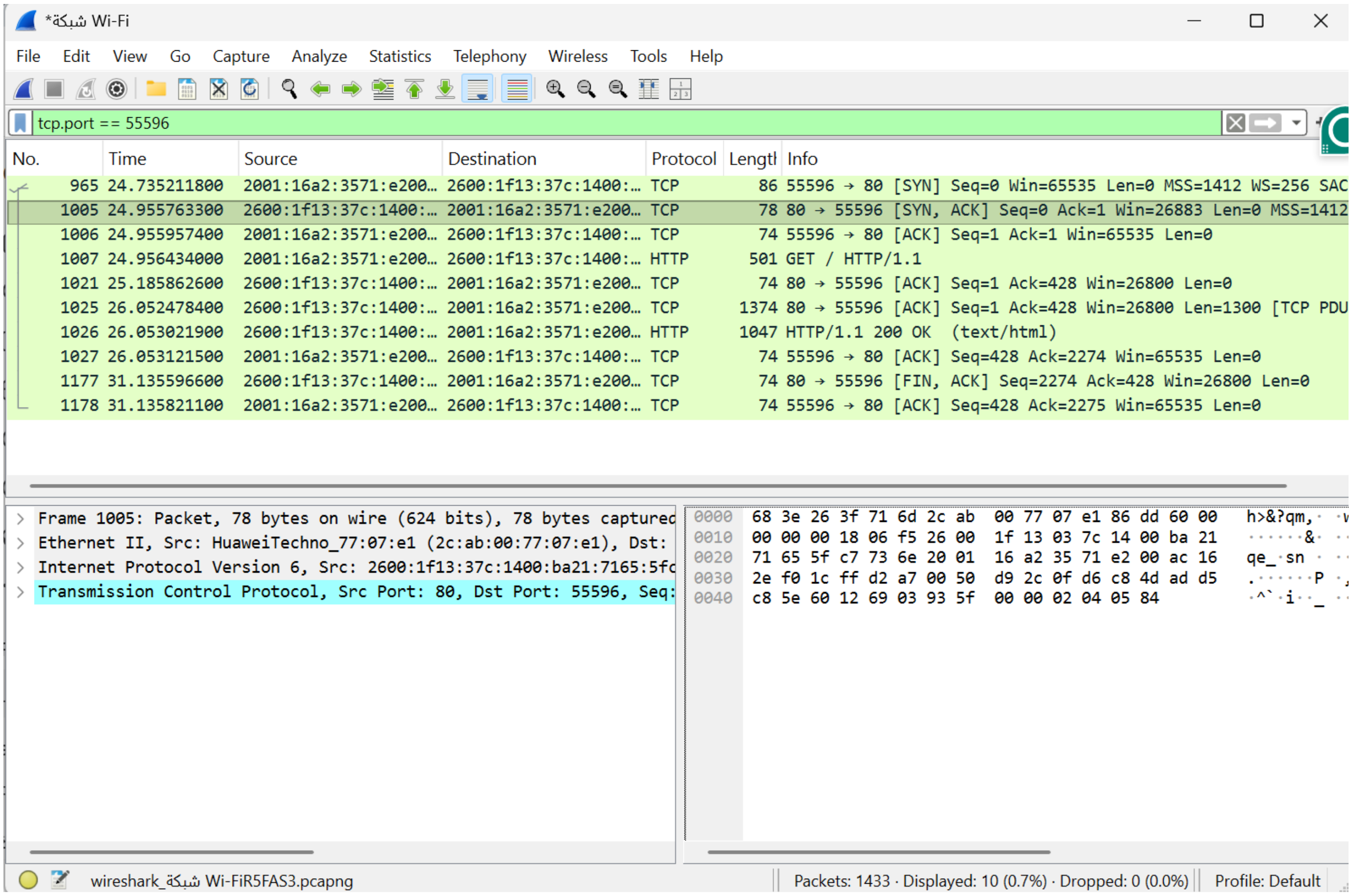
Part 1: Capturing HTTP Traffic

Screenshot 1 – HTTP Packets

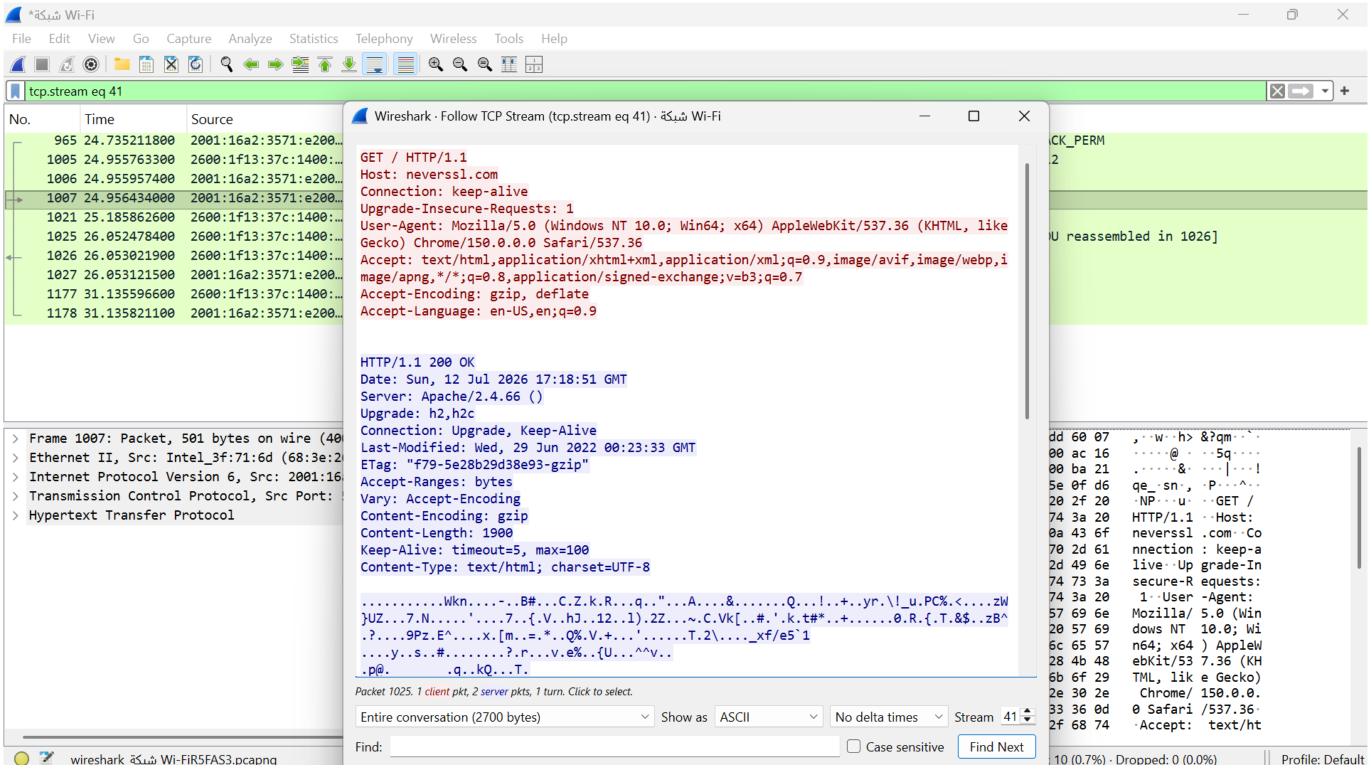


Part 2: Analyzing TCP/IP Traffic

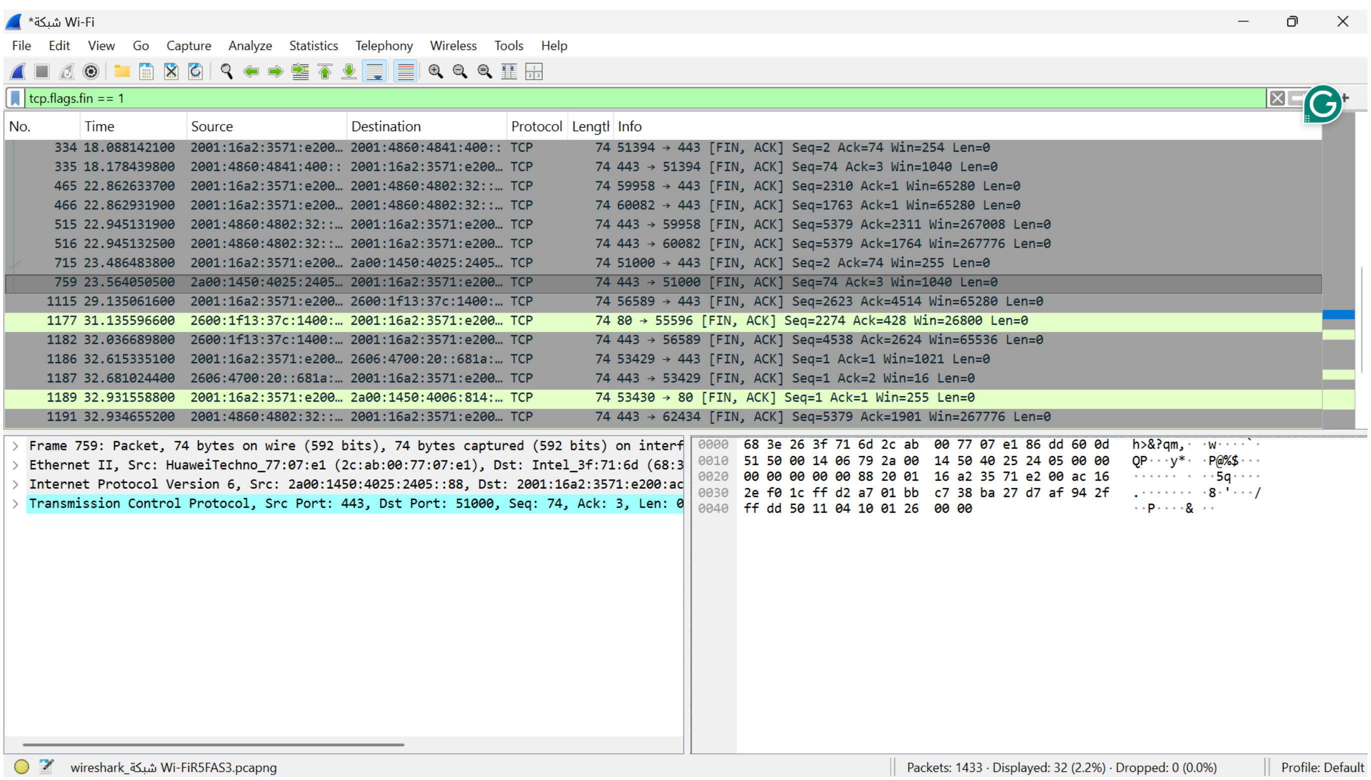
Screenshot 2 – TCP Three-Way Handshake



Screenshot 3 – TCP Data Transfer

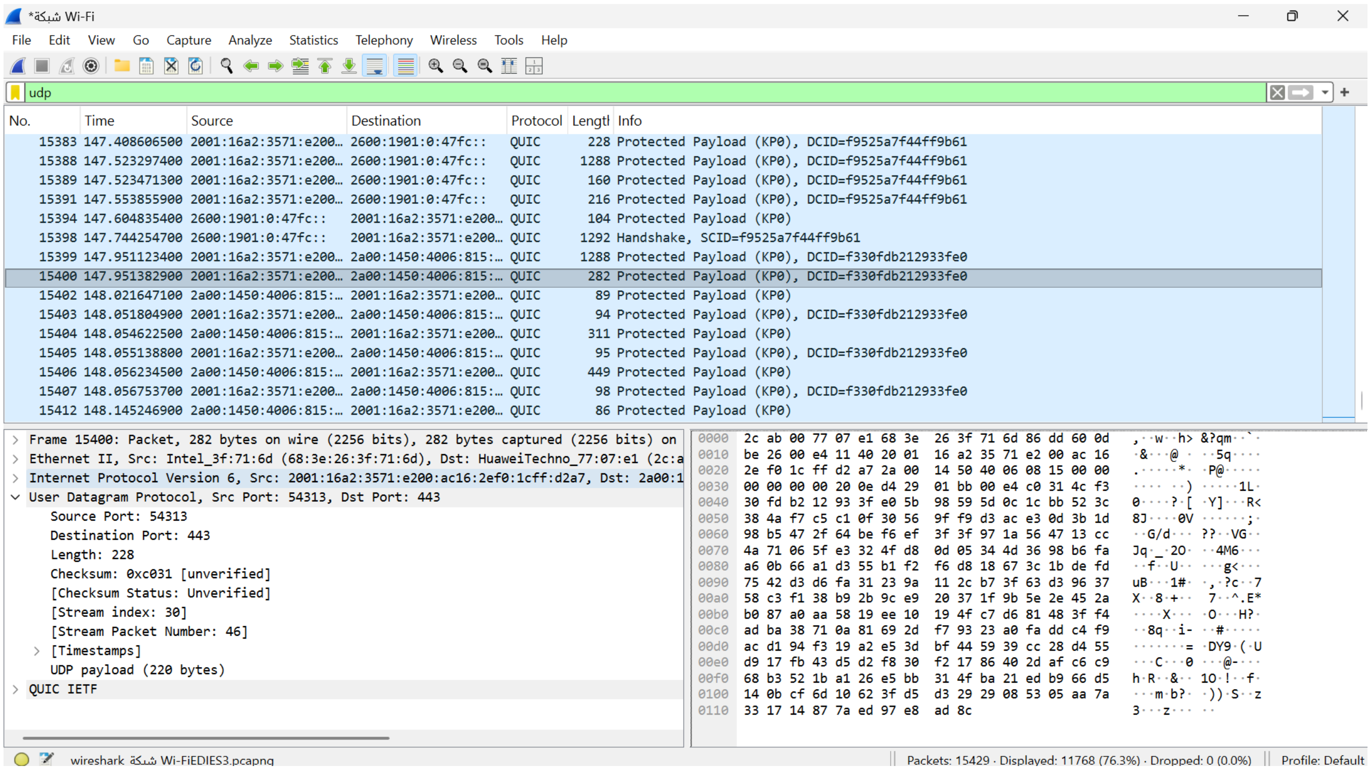


Screenshot 4 – TCP Termination



Part 3: Capturing and Analyzing UDP Traffic

Screenshot 5 – UDP Packets



Part 4: TCP and UDP Comparison

Task 1: Reliability, Connection Establishment, Data Integrity, and Ordering

Item	TCP or UDP	Reason
Reliability and Connection Establishment	TCP	TCP establishes a connection using the three-way handshake and provides acknowledgments and retransmission for reliable delivery.
Data Integrity and Ordering	TCP	TCP uses sequence numbers, acknowledgments, error checking, and retransmission to deliver data correctly and in order.

Task 2: Use Cases and Performance

	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), email, file transfer, remote login, and online banking.	Video streaming, VoIP, online gaming, DNS, and other real-time applications.
Performance	Usually slower because of connection setup, acknowledgments, retransmission, ordering, and error control.	Usually faster because it is connectionless, has a smaller header, and has less protocol overhead.